

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1**(Adv-2019-P1-Model) **Date: 25-03-2026**

Time: 09.00Am to 12.00Pm **GTA-20** **Max. Marks: 186**

25-03-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2019-P1)_GTA-20_Syllabus

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Rankers Academy JEE

Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2019-P1-Model

Time: 3:00Hr's **IMPORTANT INSTRUCTIONS** Max Marks: 186

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 5 – 12)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 13 – 18)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 22)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 23 – 30)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 31 – 36)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 40)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 41 – 48)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 49 – 54)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

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PHYSICS **Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

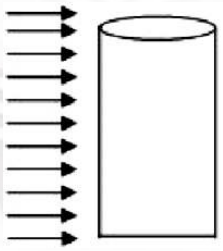
1. The radius of curvature of a concave mirror measured by a spherometer is given by $R=\frac{\ell^2}{6h}+\frac{h}{2}$. The measured values of ℓ and h are 4.0 cm and 0.060 cm respectively. The error in the measurement of R is approximately

A) 1 cm **B)** 2 cm **C)** 3 cm **D)** 4 cm

2. A double layer conducting coil is wound with no gap between two adjacent turns on a long cylindrical frame of radius R. This is called a double–layered solenoid. Diameter of cross section of the wire used is $d\left(d<<R\right)$. If current i is passed through the solenoid then the magnetic field inside it is

A) $\frac{\mu_0 i}{d}$ **B)** $\frac{2\mu_0 i}{d}$ **C)** $\frac{\mu_0 i}{2d}$ **D)** $\frac{\mu_0 i}{4d}$

3. Find the force exerted by a light beam of intensity I, incident on a cylinder (height h and base radius r) placed on a smooth surface as shown in figure if surface of cylinder is having reflection coefficient 0.8 (assume no transmission)



A) $\frac{38}{15}\frac{Irh}{c}$ **B)** $\frac{23}{15}\frac{Irh}{c}$ **C)** $\frac{70}{15}\frac{Irh}{c}$ **D)** $\frac{58}{15}\frac{Irh}{c}$

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4. It is a known fact that, the efficiency of X–ray production increases with increasing atomic number Z, it would seem that lead (Z = 82) would be a better target material than tungsten (Z = 74). However, tungsten (also used in ordinary light bulb filaments) is the most common target metal, whereas lead is never used. This is because

A) lead has too many electrons.

B) lead becomes radioactive at high temperature.

C) lead would melt due to high energy loss, whereas tungsten has a very high melting point.

D) lead will not get hot enough to produce any X–rays.

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

5. A box of volume $2V$ is divided into two halves by a thin partition. The left side contains an ideal gas at pressure P_0, temperature T and the right side is initially vacuum. A small hole of area A is punched in the partition. Because the hole is small, we can assume the gases of the two sides are at thermal equilibrium at any moment. If the number of particles (gas molecules or atoms) of the left side per unit volume at t = 0 is n_0, the numbers of particles of the left side per unit volume at the time t are $n_1(t), P_1$ is the pressure in the left side at an instant t, (average speed of gas is $v=\sqrt{\frac{8kT}{\pi m}}$, where m is the mass of each gas molecule, Assume temperature T is constant with time on both sides), then

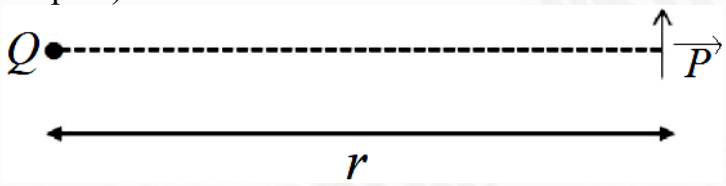
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A) $n_1(t)=\frac{n_0}{2}\left(1+e^{\left(\frac{-Avt}{2V}\right)}\right)$ **B)** $P_1(t)=\frac{P_0}{2}\left(1+e^{\left(\frac{-Avt}{2V}\right)}\right)$

C) $n_1(t)=\frac{n_0}{2}\left(1+e^{\left(\frac{-Avt}{V}\right)}\right)$ **D)** $P_1(t)=\frac{P_0}{2}\left(1+e^{\left(\frac{-Avt}{V}\right)}\right)$

6. For the situation shown in the figure (assume $r >>$ length of the dipole), select the correct statement(s). (P = dipole moment, Q = charge on the particle which is on equatorial line of dipole)



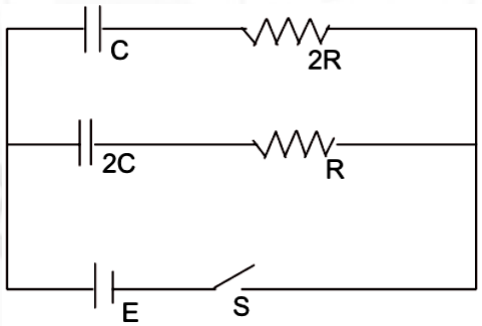
A) Force acting on the dipole is zero.

B) Force acting on the dipole is approximately $\frac{PQ}{4\pi\varepsilon_0 r^3}$ and is acting upward.

C) Torque about centre of dipole, acting on the dipole is $\frac{PQ}{4\pi\varepsilon_0 r^2}$ in clockwise direction.

D) Torque about centre of dipole, acting on the dipole is $\frac{PQ}{4\pi\varepsilon_0 r^2}$ in anti–clockwise direction.

7. In the circuit shown, switch 's' is closed at $t=0$: select the correct statement (s).



A) Rate of increase of charge is same in both the capacitors.

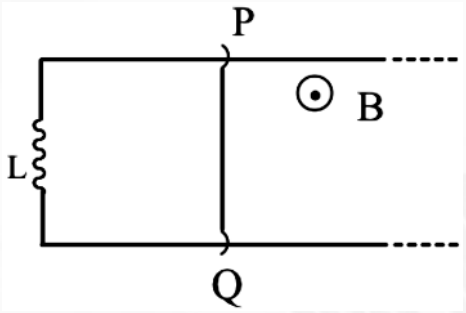
B) Ratio of charge stored in capacitors C and 2C at any time would be 1:2.

C) Time constants of both the capacitors are equal.

D) Steady state charge ratio on C and 2C is 1 : 2.

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8. A fixed U–shaped frictionless frame having an inductor of inductance L lies with its plane parallel to a horizontal plane. There exists a uniform magnetic field B throughout the region perpendicular to plane of the frame. Neglect the resistance of frame. The resistanceless slider contact PQ of length ℓ and mass m is given a velocity v_0 towards right at an instant $t=0$. Then



A) the rod executes SHM with an angular frequency $\frac{B\ell}{\sqrt{mL}}$

B) the rod executes SHM with an angular frequency $\sqrt{\frac{B\ell}{mL}}$

C) the current amplitude through the inductor due to the oscillation of the slider rod PQ is $v_0\sqrt{\frac{m}{L}}$

D) the rod executes periodic motion, not simple harmonic with time period $\frac{2\pi\sqrt{mL}}{B\ell}$

9. A chain of length L and mass per unit length ρ is piled on a horizontal surface. One end of the chain is lifted vertically with constant velocity v by a force P. Then

A) $P=\rho\left(gx+v^2\right)$, where x is length of chain in air above the surface

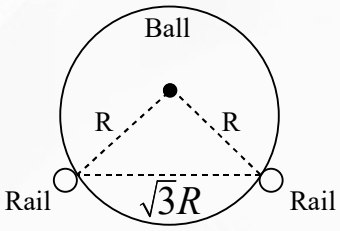
B) work done by force to completely lift the chain = $\frac{1}{2}\rho gL^2+\rho v^2L$

C) work done by force to completely lift the chain = $\frac{1}{2}\rho gL^2+\frac{\rho v^2L}{2}$

D) $P=\rho\left(gx+\frac{v^2}{2}\right)$, where x is length of chain in air above the surface

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10. A uniform solid ball of radius R rolls without slipping between the rails such that the horizontal distance is $\sqrt{3}R$ between the two contact points of the rails with the ball. Figure (a) shows the front view of the ball and figure (b) shows the side view of the ball. V_{CM} is the velocity of centre of mass of the ball and ω is the angular velocity of the ball after rolling down a distance 2h starting from rest along the incline. Then,



(Velocity into the plane)

Fig (a)

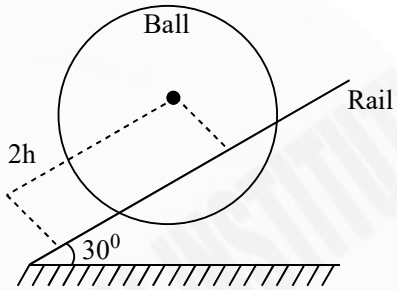
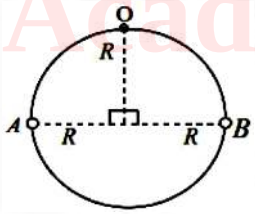


Fig (b)

A) $V_{CM} = \omega R$ **B)** $V_{CM} = \omega\dfrac{R}{2}$ **C)** $V_{CM} = \sqrt{\dfrac{10gh}{13}}$ **D)** $V_{CM} = \sqrt{\dfrac{10gh}{7}}$

11. A light rigid ring of radius R = 1 m has two identical beads A and B each of mass m = 1 kg. The ring is hinged as its top point O in such a way that it can rotate in the vertical plane freely about a horizontal axis into the plane of the diagram shown. Bead A is rigidly attached to ring and bead B is free to slide along the ring without friction. Immediately after the system is released, (Take $g = 10\,m/s^2$)



A) the acceleration of bead B has a magnitude of $\left(\sqrt{10}\right)\left(\dfrac{g}{3}\right)m/s^2$

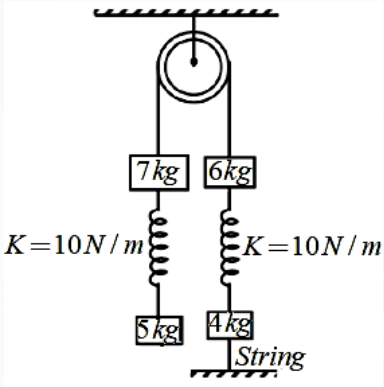
B) the ratio of magnitudes of hinge reaction at the point O along horizontal and vertical is equal to 1.

C) the acceleration of bead A is $\left(\sqrt{2}\right)\left(\dfrac{g}{3}\right)m/s^2$

D) the net reaction force from the hinge O is $\left(\sqrt{2}\right)\left(\dfrac{20}{3}\right)N$

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12. In the given diagram, pulley is frictionless and massless. Both the springs are having same force constant 10 N/m. Initially, with the string attached to the ground, the total system is at rest. Now if the string is cut, then immediately after cutting the string, $\left(g = 10 m/s^2\right)$



A) acceleration of 7 kg block is zero

B) acceleration of 5 kg block is zero

C) acceleration of 6 kg block is $\frac{100}{6} m/s^2$

D) acceleration of 4 kg block is $5\ m/s^2$

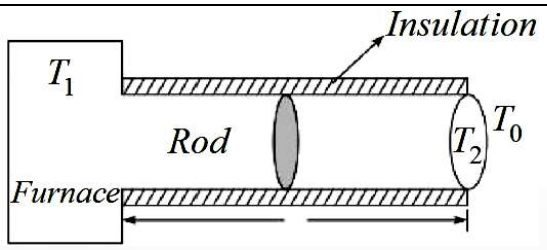
SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

13. A cylindrical rod of length L, thermal conductivity K and area of cross section A has one end in a furnace maintained at constant temperature. The other end of the rod is exposed to surrounding. The curved surface of the rod is well insulated from the surrounding. The surrounding temperature is T_0 and the furnace is maintained at $T_1 = T_0 + \Delta T_1$. The exposed end of the rod is found to be slightly warmer then the surrounding with its temperature maintained at $T_2 = T_0 + \Delta T_2 \left(\Delta T_2 << T_0\right)$. The exposed surface of the rod has emissivity 'e'. Then we can show that, in steady state, $\Delta T_1 = \left(\frac{4e\sigma L T_0^3}{K} + h\right)\Delta T_2$, where h is an integer. Find h.

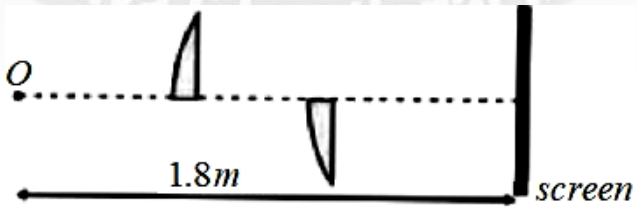
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14. A particle having specific charge σ is projected in xy plane with a speed v. There exists a uniform magnetic field in z–direction having a fixed magnitude B_0. The field is made to reverse its direction after every interval of $\frac{2\pi}{\sigma B_0}$. Calculate the maximum separation (in m) between two positions of the particle during its course of motion. (Given $\frac{v}{\sigma B_0} = 2$ m) (neglect any other force including gravity throughout the motion)

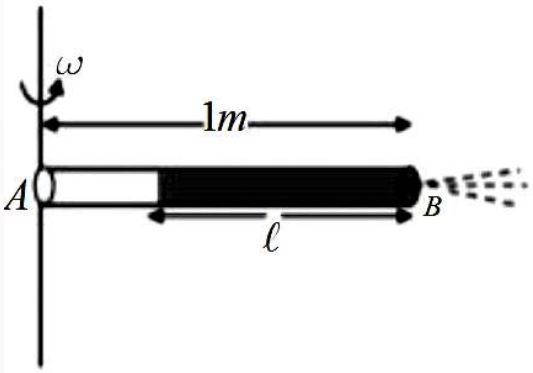
15. A parachutist jumps from the top of a very high tower with a siren of frequency 800 Hz on his back. Assume his initial velocity to be zero. After falling freely for 12s, he observes that the frequency of sound heard by him reflected from level ground below him is differing by 700 Hz w.r.t the original frequency. If the height of tower is given by (1000 + 10x + 7.5) m, then find the value of x. Velocity of sound in air is 330 m/s, and $g = 10 m/s^2$.

16. A thin plano–convex lens of focal length f is split into two halves, one of the halves is shifted along the optical axis so that both the pieces form image of 'O' on the screen. The separation between object and image plane is 1.8 m. The magnification of the image formed by one of the half lens is 2. The focal length of lens is 5n cm. Find n.

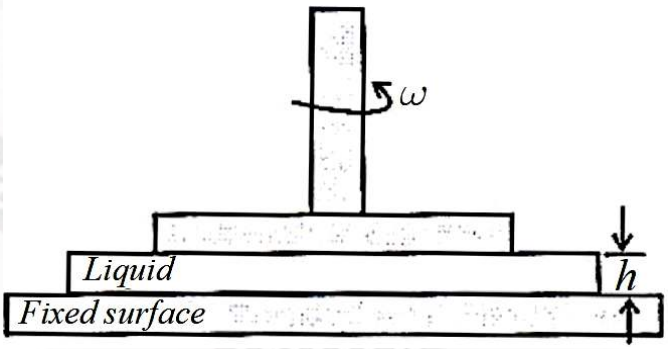


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17. A narrow horizontal tube AB of length 1 m open at A and closed at B is filled with an ideal liquid. The end B has a small orifice. The tube is set in rotation in horizontal plane with angular velocity $\omega = 10 rad/s$ about a vertical axis passing through end A. The speed of ejecting liquid with respect to the tube is V m/s and the length of the liquid column is $\ell = 0.4m$ at some instant. If the value of V at that given instant of time is $(4 \times n) m/s$, then n=?



18. A circular disc of a diameter d is slowly rotated in a liquid of large viscosity μ at a small distance h from a fixe surface. The expression for torque τ necessary to maintain an angular velocity ω is $\frac{\mu \pi d^4 \omega}{(8h \times n)}$. Find the value of n. (neglect edge effects)



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CHEMISTRY **Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

19. Four different refining techniques were used to purify four different sets of metals. Each technique relies on a distinct physical or chemical property of the metal.

1) One technique works due to the difference in solubility of impurities between the molten and solid states of the metal.

2) One technique involves the formation of a volatile compound of the metal, which decomposes on heating.

3) One technique is applicable when the metal has a boiling point much lower than that of its impurities.

4) One technique is useful when the metal has a low melting point, and the impurities are non-fusible.

The sets of metals purified are:

Set I: Ge, Si Set II: Ni, Zr, Ti

Set III: Zn, Hg Set IV: Sn

Which of the following is the correct match for refining of metals?

A) 1 – Set I, 2 – Set II, 3 – Set III, 4 – Set IV

B) 1 – Set II, 2 – Set I, 3 – Set IV, 4 – Set III

C) 1 – Set III, 2 – Set IV, 3 – Set I, 4 – Set II

D) 1 – Set IV, 2 – Set III, 3 – Set II, 4 – Set I

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20. o–Nitrotoluene is subjected to the following sequence of reactions as shown. Assume that A to F are the major organic products formed at each step.

$$o-Nitrotoluene \xrightarrow{Sn/HCl} (A)$$

$$(A) \xrightarrow{Acetic\ anhydride} (B)$$

$$(B) \xrightarrow{conc.HNO_3/conc.H_2SO_4} (C)$$

$$(C) \xrightarrow{KOH(aq.)} (D)$$

$$(D) \xrightarrow{Cl_2(1equiv.),350^\circ C} (E)$$

$$(E) \xrightarrow[2.Cu/HCl,\Delta]{1.NaNO_2/HCl,0^\circ-5^\circ C} (F)$$

Identify the only INCORRECT statement regarding the above sequence of reactions.

A) Both (A) and (E) form a foul-smelling compound on reaction with $CHCl_3$ and KOH.

B) Compound (A) is more basic than compound (B).

C) One mole of compound (F), on heating with excess aqueous $AgNO_3$, forms two moles of white precipitate which readily dissolves in aqueous NH_3.

D) Both (E) and (F), on reaction with Zn / NH_4Cl followed by treatment with Tollens' reagent, give a silver mirror.

21. A first-order gas-phase decomposition occurs in a rigid vessel:

$X(g) \rightarrow 2Y(g) + Z(g)$ (first order in X)

Initially, 3.00 mol of $X(g)$ is taken in a 10.0 L rigid, closed container at 300 K. After 10 s, the reaction is quenched instantaneously. Immediately, a tiny pinhole is made in the vessel (temperature remains 300 K) so that the gases begin to effuse out.

After the pinhole is made, the initial molar ratio of effusing gases X : Y is observed to be 1 : 2. Assume ideal gas behavior and that Graham's law applies.

Given: M(X)=16 $g\,mol^{-1}$, M(Y)=4 $g\,mol^{-1}$, M(Z)=8 $g\,mol^{-1}$; ln2 = 0.7, ln3 = 1.1;

R = 0.08 $atm.L.mol^{-1}K^{-1}$.

Which of the following statements is INCORRECT?

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A) The total pressure inside the vessel just after quenching is 12 atm.

B) The rate constant, k, equals $4\times10^{-2}s^{-1}$.

C) The amount of Z present in the vessel just after quenching is 1.0 mol.

D) The rate of formation of Y at t = 10 s is 0.4 $mol\,L^{-1}s^{-1}$.

22.

$$\underset{\text{(salt mixture)}}{A} \xrightarrow{\text{excess NaOH}} P\downarrow + Q\text{ (solution)}$$

$$Q\text{ (solution)} \xrightarrow[\text{(iii) } K_2Cr_2O_7/\text{conc.}H_2SO_4]{\text{(i) dil. HCl (ii) filter and dry residue}} R\uparrow\text{ (red fumes)}$$

$$R \xrightarrow{NaOH} S\text{ (yellow solution)}$$

$$S \xrightarrow{AgNO_3} T\downarrow\text{ (brick-red ppt)}$$

$$P \xrightarrow{\text{(i) conc. HCl (ii) KI}} U\downarrow\text{ (black ppt)}$$

$$U \xrightarrow{\text{excess KI}} V\text{ (orange soluble complex)}$$

Salt mixture 'A' is subjected to the qualitative reactions shown in the above reaction scheme. Based on the observations, answer the following question.

Which of the following statements is/are correct?

1) R is chromyl chloride, CrO_2Cl_2. 2) T is silver chromate, Ag_2CrO_4.

3) U is bismuth(III) iodide, BiI_3. 4) P is bismuth(III) hydroxide, $Bi(OH)_3$.

A) 1, 2, 4 B) 2, 3 **C)** 1, 2, 3, 4 **D)** 1, 3, 4

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

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23. In the following transformation network,

$$HIO_3 \underset{2}{\overset{1}{\rightleftharpoons}} A \xrightarrow[5]{3} ICl_3 \xrightarrow{4} Ba(IO_3)_2 \xrightarrow{6} HIO_3 \text{ and also } A \xrightarrow{5} Ba(IO_3)_2.$$

'A' is iodine. Choose all correct options.

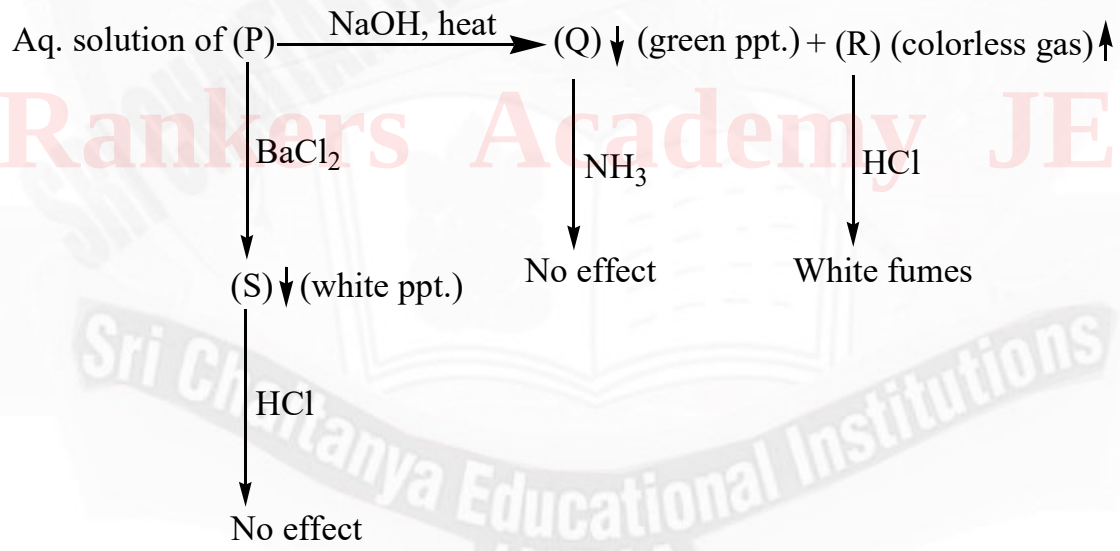
A) Step (3) can be carried out by chlorine in an inert solvent such as CCl_4 represented by $I_2 + 3Cl_2 \rightarrow 2ICl_3$.

B) Step (1) represents the oxidation of iodide by iodic acid, leading to liberation of iodine represented by $HIO_3 + 5HI \rightarrow 3I_2 + 3H_2O$.

C) Step (6) is driven by precipitation of insoluble barium sulfate using dilute sulfuric acid represented by $Ba(IO_3)_2 + H_2SO_4 \rightarrow BaSO_4 \downarrow + 2HIO_3$.

D) In step (4), treatment of ICl_3 with $Ba(OH)_2$ leads to formation of $Ba(IO_3)_2$ along with $BaCl_2$ and BaI_2.

24.



Consider the reaction scheme shown above. An aqueous solution of (P) is subjected to the given tests. Based on the observations, which of the following statement(s) is/are correct?

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A) (P) can give a brown precipitate with alkaline $K_2[HgI_4]$.

B) (Q) can turn brown on exposure to air.

C) (P) can give a blue precipitate with $K_4[Fe(CN)_6]$ in the absence of air.

D) (P) can give a red coloured solution with NH_4SCN in absence of air.

25. The following compounds are strongly heated under identical conditions:

i) KNO_3 ii) $Cu(NO_3)_2$ iii) $Pb(NO_3)_2$ iv) NH_4NO_3

Which of the following statement(s) regarding the gaseous products formed is/are correct?

A) Compounds (ii) and (iii) liberate colourless paramagnetic gas whose magnetic moment can be taken to zero by cooling.

B) Compound (iv) liberates a neutral oxide in terms of acid-base properties.

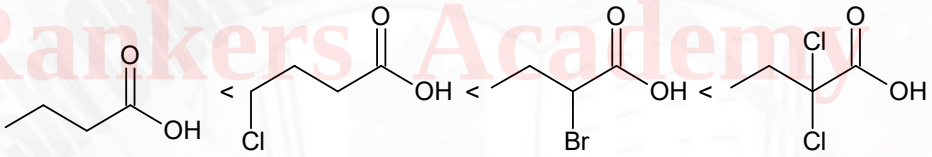
C) All compounds (i), (ii), (iii) and (iv) liberate paramagnetic gases.

D) Compounds (i), (ii) and (iii) liberate paramagnetic homodiatomic molecule

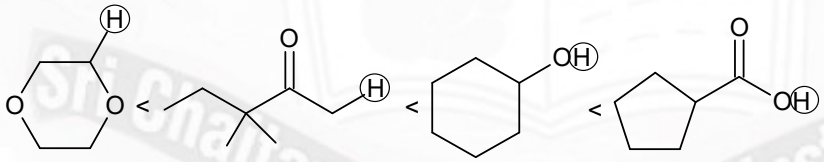
26. Which of following represent the correct increasing order of acidity?

Acidic strength is to be compared for indicated proton by circle; if no proton is marked, compare the most acidic proton in that molecule.

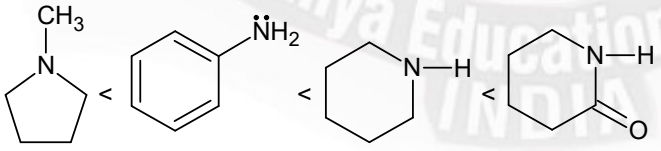
A)



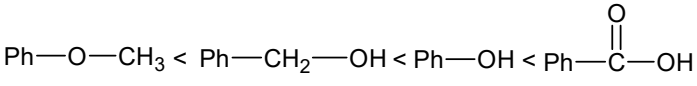
B)



C)



D)



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27. The molar conductivity of a 0.012 M aqueous solution of chloroacetic acid $(C\ell - CH_2 - COOH)$ is $100\Omega^{-1}cm^2mol^{-1}$. The ionic molar conductivity at infinite dilution of $C\ell - CH_2 - COO^-$ and H^+ ions are 50 and $350\Omega^{-1}cm^2mol^{-1}$ respectively. (Given: log 3 = 0.48). Which of the following statements is/are correct?

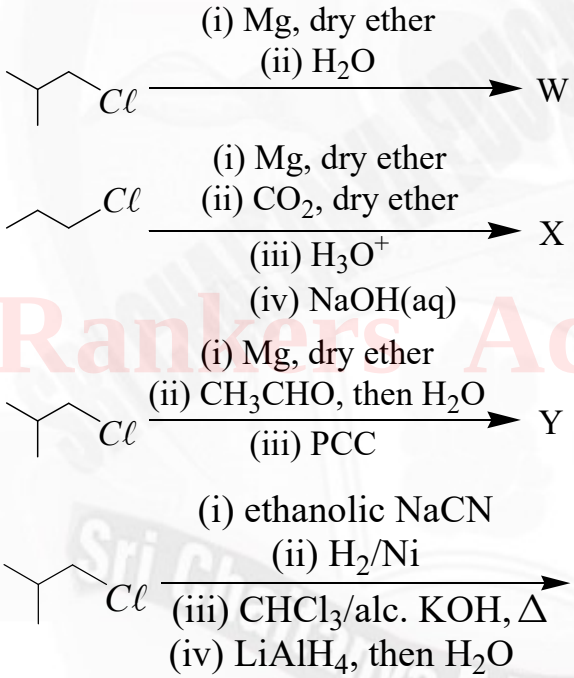
A) The degree of dissociation of the acid is 0.25.

B) The hydrogen ion concentration is $3\times10^{-3}M$.

C) The pK_a of $C\ell - CH_2 - COOH$ is 3.

D) The pH of the solution is 2.52.

28. In the following reaction sequences, W, X, Y and Z are the major organic products formed:



The correct statement about W, X, Y and Z is:

A) Y contains six carbon atoms and undergoes the Cannizzaro reaction.

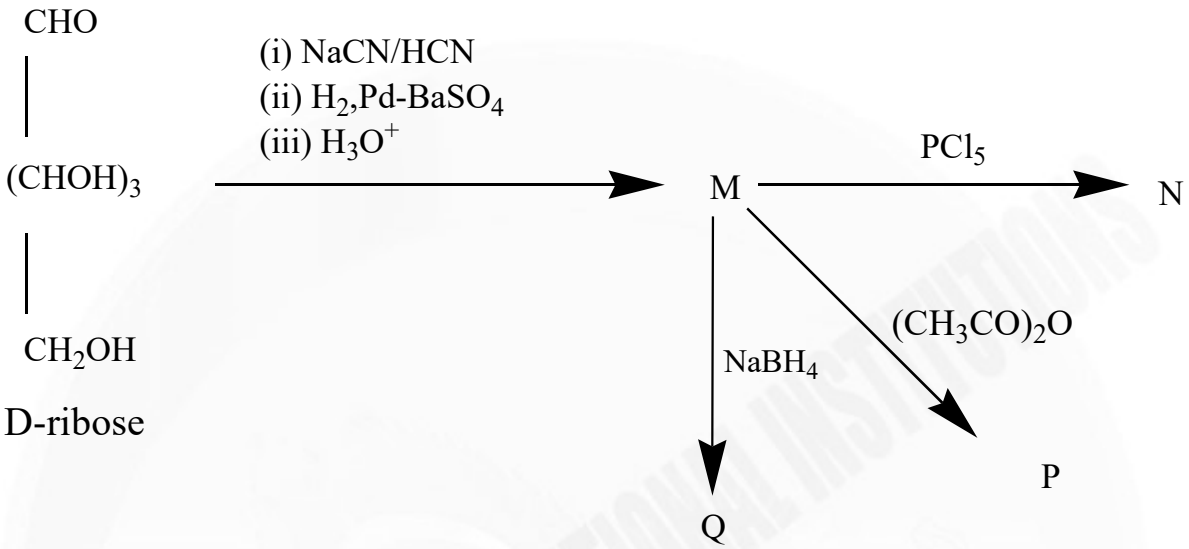
B) W is a four carbon primary alcohol.

C) Z is a six carbon primary amine.

D) X undergoes Kolbes electrolysis to give a six–carbon alkane.

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29. In the following sequence of reactions, M, N, P and Q are the major products formed:



Which of the following options is/are correct?

A) M is a enantiomeric mixture with formula $CHO(CHOH)_4CH_2OH$

B) N is a dichloride

C) P is a pentaacetate

D) Q is a diastereomeric mixture with formula $CH_2OH(CHOH)_4CH_2OH$

30. For a hydrogen atom described using the Bohr atomic model, consider an electron moving from the ground state to the first excited state.

Which of the following statements correctly describe the relative magnitudes of physical quantities associated with the electron in these two states?

A) The electron experiences a larger centripetal acceleration in the first excited state than in the ground state.

B) The quantized angular momentum of the electron increases on going from the ground state to the first excited state.

C) The kinetic energy of the electron decreases when it moves from the ground state to the first excited state.

D) The electrostatic force acting on the electron in the first excited state is stronger than that in the ground state.

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SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

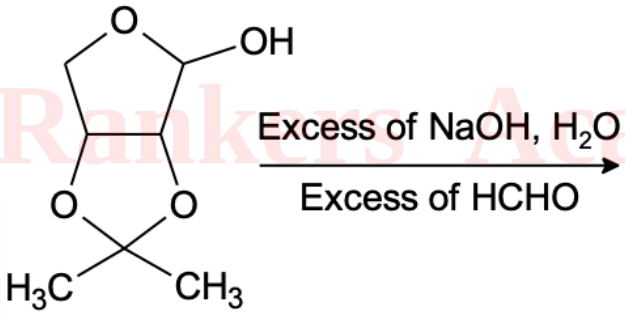
- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

31. 200 mL of a saturated aqueous solution of $CaC_2O_4(s)$ requires 8.0 mL of $5.0\times10^{-3}N$ $KMnO_4$ solution for complete oxidation in acidic medium. What is value of $-\log K_{sp}$ for $CaC_2O_4(s)$?

32. An electron in a He^+ ion is in a state having directional dependence and a total of one node. A photon of frequency (5RC/9) causes a transition to a higher state that can have five possible Z–components for same orbital angular momentum. What is the magnitude of change in $(n-\ell)$?

(R = Rydberg constant, C = speed of light)

33.



Number of hetero atoms in the major product of the above reaction is _____

34. A tranquilizer T is synthesized according to the following reaction scheme. The molar mass of the tranquilizer T is _____ g/mol.

(Use C = 12, H = 1, N = 14, O = 16 g/mol)

$$C_8H_{16} \xrightarrow[Zn/H_2O]{O_3} 2Q \xrightarrow[\Delta]{NaOH} R \xrightarrow{Ag_2O/NH_4OH} S \xrightarrow[(ii)NH_3]{(i)SOCl_2} T$$

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35. An element E has the ground-state valence electronic configuration ns^2np^x. It forms a stable fluoride of formula EF_Y, in which the element E attains the excited-state configuration $ns^2np^{(x-2)}nd^2$. The molecule EF_Y contains a total of 16 lone pairs. Find the value of (x + y).

36. When a neutral organic compound X is added to water and ethyl acetate (organic solvent) mixture the following physical equilibrium will be established:

$X_{(aq)} \rightleftharpoons X_{(org)}$, equilibrium constant, $K_D = \dfrac{[X]_{org}}{[X]_{aq}} = 4$ (At experimental temperature)

An aqueous solution of volume 100 mL containing X is extracted three times, each time with 100 mL of fresh ethyl acetate. After each extraction, the organic layer is separated and replaced by fresh ethylacetate. Assume volumes remain unchanged and equilibrium is established in each extraction.

What is the percentage of X present in the combined organic phase after the three extractions ?

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MATHEMATICS **Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

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37. Let $P(x)=24x^{24}+\sum_{j=1}^{23}(24-j)\left(x^{24-j}+x^{24+j}\right)$. Let $x_1,x_2.....x_r$ be the distinct zeros of P(x) and let ${x_k}^2=a_k+ib_k$ for k = 1 2, ... r where a_k,b_k are real numbers, and $i^2=-1$.

Let $\sum_{k=1}^{r}|b_k|=m+n\sqrt{p}$ where m, n and p are integers and p is prime then the value of m + n + p is equal to _____

A) 7 **B)** 12 **C)** 11 **D)** 15

38. The value of $\lim_{x\to 0}\left(\int_0^1\left(by+a(1-y)\right)^x dy\right)^{\frac{1}{x}}$ where $b>a$ is

A) $\frac{1}{e}\left(\frac{b^b}{a^a}\right)^{\frac{1}{b-a}}$ **B)** $\frac{b^a}{a^a}$ **C)** $\left(\frac{b^b}{a^a}\right)^{\frac{1}{b-a}}$ **D)** $e\left(\frac{b^b}{a^a}\right)$

39. Let $a_0=\frac{17}{4}$ and $\frac{a_k+2}{a_{k-1}}=a_{k-1}$ for $k\geq 1$ then the value of $\left\lfloor 7.\prod_{k=0}^{\infty}\left(1-\frac{1}{a_k}\right)\right\rfloor$ is equal to ___

A) 4 **B)** 9 **C)** 5 **D)** 8

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40. Let $f(x)=\lim\limits_{n\to\infty}\left[\dfrac{n^n(x+n)\left(x+\frac{n}{2}\right)....\left(x+\frac{n}{n}\right)}{n!\left(x^2+n^2\right)\left(x^2+\frac{n^2}{4}\right)...\left(x^2+\frac{n^2}{n^2}\right)}\right]^{\frac{x}{n}}$, for all x > 0 then which is

INCORRECT?

A) $f\left(\frac{1}{4}\right)\le f\left(\frac{2}{5}\right)$ **B)** $f\left(\frac{1}{6}\right)\ge f\left(\frac{2}{3}\right)$ **C)** $\frac{f'(5)}{f(5)}\le\frac{f'(2)}{f(2)}$ **D)** $f'(4)\le 0$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

41. Let $f:(0,\infty)\to\mathbb{R}$, has an anti-derivative g such that $f(x).g\left(\frac{1}{x}\right)=\frac{1}{x}\forall x>0$ and $f(1)=2$. Then which of the following is/are **CORRECT**? ([.] denotes GIF)

A) $\left[f\left(\frac{1}{3}\right)\right]=0$ **B)** $\left[\sqrt[4]{f(2)}\right]=2$ **C)** $\left[f(2)^{\frac{1}{3}}\right]=2$ **D)** $\left[f(4)\right]=14$

42. Let $S_k=\sum\limits_{j=0}^{k}2^{k-j}\binom{k+j}{j};k\in W$, ($\binom{n}{r}$ represents ${}^nC_r$) then which of the following is/are **CORRECT**?

A) $S_{2026}=4S_{2025}$ **B)** $S_{2025}=16S_{2023}$

C) $\sum\limits_{k=1}^{2026}S_k=\frac{2}{3}\left(2^{4052}-1\right)$ **D)** $\sum\limits_{k=1}^{2025}S_k=\frac{4}{3}\left(2^{4050}-1\right)$

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43. The number of ways to colour 3cells in a 3×3 grid so that no two colored cells share an edge is R, then R is divisible by

A) 2 **B)** 3 **C)** 7 **D)** 11

44. Let $A(Z_1), B(Z_2), C(Z_3)$ are the vertices of triangle ABC respectively in anticlockwise sense such that $Arg\left(\frac{Z_2-Z_1}{Z_2-Z_3}\right)=\frac{\pi}{2}$. Z_4 be the point of intersection of $Arg\left(\frac{Z_1-Z}{Z_1-Z_2}\right)=\frac{A}{2}$ and $Arg\left(\frac{Z_3-Z}{Z_3-Z_1}\right)=\frac{C}{2}$ where A and C are the internal angles of triangle at vertices $A(Z_1)$ and $C(Z_3)$. Z_5 be the point of intersection of $Arg\left(\frac{Z_4-Z}{Z_4-Z_1}\right)=\frac{\pi}{2}$ and $Arg\left(\frac{Z-Z_2}{Z-Z_3}\right)=0$, then which of the following is/are **CORRECT?**

A) $Arg\left(\frac{Z_3-Z_4}{Z_1-Z_5}\right)=\frac{\pi}{2}$ **B)** $Arg\left(\frac{Z_3-Z_4}{Z_1-Z_5}\right)=\frac{-\pi}{2}$

C) $|Z_4-Z_5|^2=\frac{|Z_1-Z_5|^2}{2}$ **D)** $|Z_4-Z_5|=\sqrt{|Z_1-Z_3|^2-|Z_1-Z_3||Z_2-Z_3|}$

45. Which of the following statements is/are **TRUE**?

A) $2\left(\tan^{-1}\pi-\tan^{-1}e\right)-ln\pi+1<0$.

B) $\lim\limits_{n\to\infty}\sum\limits_{k=0}^{n}\frac{{}^nC_k}{n.2^n+k}=0$

C) The first group of two samples has 100 items with mean 15 and Standard deviation 3. If the combined sample of two groups has 250 items with mean 15.6 and standard deviation $\sqrt{13.44}$, then the standard deviation of the second group is 4.

D) If the mean deviation of the numbers 1, 1 + d, 1 + 2d …… 1 + 100d from their mean is 255, then d is equal to 10.2.

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46. Let P(0, 14) and Q(0, – 6) be distinct points on parabola 'C' with focus O(0, 0). Let 'L' and 'a' denotes the length of latusrectum and length of intercept on x–axis by parabola C respectively. Then which of the following is/are **TRUE**?

A) The value of L is $\frac{48}{5}$ **B)** The value of 'a' is $\frac{420}{13}$

C) The value of L is $\frac{84}{5}$ **D)** The value of 'a' is $\frac{428}{13}$

47. A differentiable function $f:R\to R$ satisfies the functional equation $f(x).f(y)+f(x+y)=e^x f(y)+e^y f(x)+xy\forall x,y\in R$. If $f'(0)=0$ and $f(0)=0$, then which of the following statements is/are **CORRECT**?

A) $\lim\limits_{x\to 0}\frac{f(x)}{x^2}=\frac{1}{2}$

B) $\int\limits_{x}^{x^2}\left(f'(t)-f(t)\right)dt>0\forall|x|>1$

C) $F(x_2)>F(x_1)\forall x_2>x_1$, where $F(x)=f'(x)-f(x)$

D) There exists at least two horizontal tangents to the curve $y=f(x)$ in $(-1,1)$.

48. If $f(\alpha)=\lim\limits_{x\to 2}\left(\sin^x\alpha+\cos^x\alpha\right)^{\frac{1}{(x-2)}}$ for $\alpha\in\left[0,\frac{\pi}{2}\right]$, then which of the following is/are **CORRECT** ?

A) $f(0)=1$

B) $f\left(\frac{\pi}{2}\right)=1$

C) $f(\alpha)=(\cos\alpha)^{\cos^2\alpha}.(\sin\alpha)^{\sin^2\alpha}$ if $\alpha\in\left(0,\frac{\pi}{2}\right)$

D) $f(\alpha)=\frac{(\sin\alpha)^{\sin^2\alpha}}{(\cos\alpha)^{\cos^2\alpha}}$ if $\alpha\in\left(0,\frac{\pi}{2}\right)$

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SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

49. If $\text{Tan}\left(\frac{\pi}{4}+\frac{1}{2}\cos^{-1}x\right)+\text{Tan}\left(\frac{\pi}{4}-\frac{1}{2}\cos^{-1}x\right)=1$,then the number of solutions of the given equation in $(0,1)$ is _____

50. Let $A=\{2,3,4,5,6,7,8\}$ and $B=\{3,6,7,9\}$. The number of elements in the set $\{C\subseteq A: C\cap B\neq\phi \text{ and } 8\in C\}$ is R then the largest digit in R is _____

51. Number of ordered triplets $(x,y,z), x,y,z\in R$ such that $81^{x^2+y}+81^{y^2+z}+81^{z^2+x}=1$ is _____

52. Let $A=\left[a_{ij}\right]$ be a real matrix of order 3×3, such that $a_{i1}+a_{i2}+a_{i3}=2$, for i=1,2,3. Then, the sum of all the entries of the matrix A^3 is equal to :

53. Let X be the set consisting of the first 2026 terms of arithmetic progression 2, 7, 12, … and Y be the set consisting of the first 2026 terms of arithmetic progression 3, 10, 17, 24, …… Then number of elements in the set $X\cap Y$ is _____

54. If the graphs $y=3(x-\alpha)^2+\beta$ and $y=2(x-\alpha)^2+\gamma$ have y-intercepts of 2013 and 2014 respectively, and each graph has two positive integer x-intercepts $(\alpha,\beta,\gamma\in R)$. Then the value of $\left(\frac{\alpha}{\gamma-\beta-1}\right)^{-1}$ is equal to _____

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16 RANK DEVDUTTA MAJHI HT. No. 255053116*	18 RANK DHARMANA GNANA RUTVIK SAI HT. No. 256055278	19 RANK VANGALA AJAY REDDY HT. No. 256131009	23 RANK AKSH GOGI HT. No. 252071075*	26 RANK P HEMA SAI SURYA KARTHIK HT. No. 256033006	27 RANK SARKARSAMUDRA HT. No. 252071105*
30 RANK OM PRAKASH BEHERA HT. No. 252021018*	32 RANK SUNKARA SAI RISHANTH REDDY HT. No. 256165327	34 RANK DHRUBA JYOTHI PANJA HT. No. 252048246*	35 RANK BHAVESH JAYANTHI HT. No. 251043080	36 RANK ADVAY MAYANK HT. No. 252104113*	37 RANK KARMANYA GUPTA HT. No. 252081477*
42 RANK MD ANAS HT. No. 252046210*	45 RANK RAMIT GOYAL HT. No. 257001113*	52 RANK MAULIK JAIN HT. No. 252079407*	54 RANK GARV HT. No. 252056188*	59 RANK LARISSA HT. No. 252079071*	60 RANK ARYAN BALABADRULA HT. No. 256132077
63 RANK SAMYAJYOTI BISWAS HT. No. 255038455*	64 RANK AARUSH ANAND HT. No. 251006176*	72 RANK RUSHMITH BANDARI HT. No. 256168048	78 RANK KORIKANA RASAGNYA HT. No. 256057046	87 RANK LAKSHYA SHARMA HT. No. 252070079*	91 RANK AVANEESH BANSAL HT. No. 251113130*	95 RANK KAVYA AGGARWAL HT. No. 252079121*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS: 29	BELOW 500 ALL INDIA OPEN CATEGORY RANKS: 113	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS: 205	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT: 745	NUMBER OF QUALIFIED RANKS: 4,212

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